Machine Learning for Operational Decisions

From a demand forecast to an inventory decision

Automation and Robotics

What should a forecast help us decide?

A robotic cell prepares kits before tomorrow's orders arrive.

- Too few kits: missed orders or expensive emergency assembly.
- Too many kits: wasted preparation and storage.
- A demand forecast describes what may happen. An inventory policy chooses what to do.

By the end, you should be able to

Define the decision and its costs; explain why low forecast error need not mean low operational cost; evaluate a policy on future data without leakage.

All numerical examples in this lecture are invented teaching examples.

Prediction, optimization, and feedback

$$\underbrace{\text{past data} + \text{information available now}}_{x_t} \longrightarrow \underbrace{\text{forecast of tomorrow}}_{\widehat{P}(D_t|x_t)} \longrightarrow \underbrace{\text{choose } q_t}_{\text{decision}}$$

- ML estimates uncertain inputs: demand, travel times, failure risk.
- OR chooses feasible actions using an objective: cost, delay, or service.
- After demand is observed, measure forecast error *and* realized decision cost.
- Compare with a simple baseline. Complexity alone does not establish value.

Ask: Which information exists when the cell commits to tomorrow's quantity?

A model starts with assumptions

- One product, one day; choose integer quantity $q \geq 0$ before demand D is known.
- Prepared kits are available before sales; no additional replenishment that day.
- Unused preparation has no value tomorrow; unmet demand is lost.
- No starting inventory, setup charge, or shared capacity in the first model.
- Demand distribution is unaffected by our quantity or price; demand is observable even when we run out.

Limits matter

This is a single-period newsvendor model. Reusable stock, lead times, multiple products, or censored demand require additional modeling.

Define the loss before selecting the predictor

Let $c_o > 0$ be excess cost per unused kit and $c_u > 0$ shortage cost per unserved kit.

$$L(q, D) = c_o (q - D)_+ + c_u (D - q)_+ , \qquad (z)_+ = \max(z, 0).$$

- q, D: kits; c_o, c_u: currency/kit; L: currency/day.
- Example: $c_o = 1$, $c_u = 9$, $q = 100$, $D = 120$ gives $L = 180$.
- L measures mismatch cost; it is not total purchasing expenditure.

For purchase cost w, sale price p, salvage s ($p > w > s$), take $c_u = p - w$, $c_o = w - s$. Then realized profit is $(p - w)D - L(q, D)$; maximizing expected profit is equivalent if demand is exogenous.

Three possible demand outcomes

Assume $c_o = 1$, $c_u = 9$, with this known distribution:

Demand D	80	100	120
Probability	0.25	0.50	0.25
$L(100, D)$	20	0	180
$L(120, D)$	40	20	0

$$\mathbb{E}[L(100, D)] = 50, \qquad \mathbb{E}[L(120, D)] = 20.$$

Predict before calculating: Why might preparing more than mean demand 100 be worthwhile? What changes if shortage costs fall?

The optimization problem

With scenario values d_s and probabilities p_s:

$$\min_{q \in \mathbb{Z}_{\geq 0}} \mathbb{E}[L(q, D)] = \min_{q \in \mathbb{Z}_{\geq 0}} \sum_s p_s \left[c_o \left(q - d_s \right)_+ + c_u \left(d_s - q \right)_+ \right].$$

- A point forecast $\hat{d}$ used as certain demand gives $q = \hat{d}$ (if feasible/integer).
- This discards uncertainty and the asymmetric consequence of errors.
- For this small example, enumerate candidate quantities; no sophisticated solver is needed.
- With capacity K, impose $q \leq K$. The unconstrained quantity may be infeasible.

Why a quantile is the right target

For integer demand, one extra kit changes expected cost by

$$\mathbb{E}[L(q+1, D) - L(q, D)] = c_o \Pr(D \le q) - c_u \Pr(D \ge q+1).$$

Since $\Pr(D \ge q+1) = 1 - F(q)$, order more while

$$F(q) < \alpha, \qquad \alpha = \frac{c_u}{c_u + c_o}.$$

A smallest optimal integer quantity

$q^* = \min\{q \in \mathbb{Z}_{\ge 0} : F(q) \ge \alpha\}$. Ties may yield several optima. For continuous demand, use the corresponding distribution quantile.

With $c_u = 9$, $c_o = 1$, target the 90th percentile: $q^* = 120$.

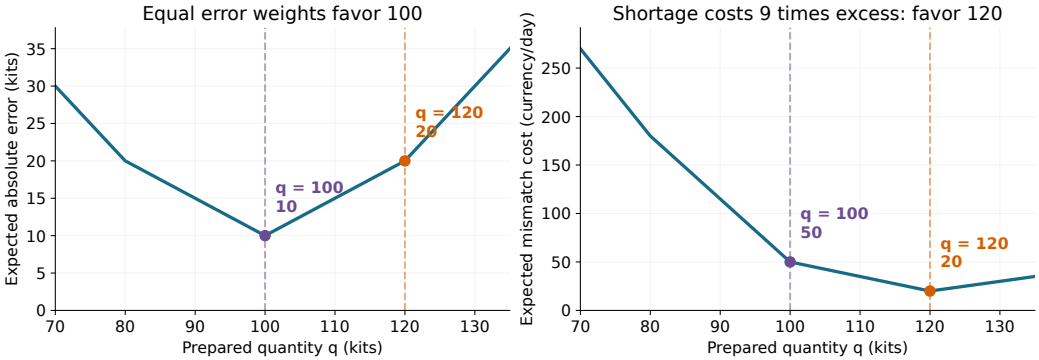
Forecast accuracy can rank policies differently

Use the same demand distribution and compare two constant outputs:

Output used as quantity	100	120
Expected absolute error (kits)	10	20
Expected mismatch cost (currency/day)	50	20
Probability demand exceeds quantity	0.25	0

- Output 100 is better for absolute error; output 120 is better for this cost.
- A median targets absolute error; a cost-weighted quantile targets this decision.
- These outputs answer different questions. Evaluate each against its intended target.

See the ranking reversal



Same demand: 80, 100, 120 with probabilities 0.25, 0.50, 0.25.

A quantity of 100 minimizes absolute error; 120 minimizes cost when $c_u = 9$, $c_o = 1$.

Service level is not a synonym for cost or fill rate

- Stockout-day rate: $\Pr(D > q)$. Cycle service level: $\Pr(D \le q)$.
- Unit fill rate: $1 - \mathbb{E}[(D-q)_+]/\mathbb{E}[D]$ when mean demand is positive.
- For $q = 100$: cycle service $= 75\%$, but fill rate $= 1 - 5/100 = 95\%$.
- With discrete demand, a target quantile can overshoot its nominal probability: $F(120) = 1$, although $\alpha = 0.9$.

Discuss: A manager asks for "95% service." Which definition do you need before modeling a constraint?

Use context to forecast the relevant uncertainty

$$q_t^* = \widehat{F}^{-1}_{D_t|x_t}(\alpha), \qquad \alpha = \frac{c_u}{c_u + c_o}.$$

- Inputs might include weekday, confirmed promotions, and previous demand.
- A first implementation: fit a point forecast, then add a quantile of past forecast residuals.
- The residual distribution must represent the decision horizon and operating regime.
- A single residual offset assumes similar error distributions across contexts. Check errors by weekday/product; changing uncertainty needs a richer model.

Uncertainty is an input to the decision, rather than an error bar to ignore.

Preserve the order of information

Training	Calibration	Validation	Test
Earlier history	Later block	Later block	Final block
Fit predictor	Fit error quantile	Select policy	Evaluate once

- Keep every test target later than all data used to fit or select its policy.
- Reproduce the actual forecast horizon. Rolling-origin validation repeats this at several cutoffs.
- Fit imputation, scaling, and feature choices using only data available at each cutoff.
- Validation selects a policy; the final test estimates performance after selection. Do not keep tuning on the test set.

The companion uses four small blocks to make these roles visible.

Three leakage and measurement traps

1. **Future features:** tomorrow's realized sales or temperature are unavailable tonight. Use lagged observations or forecasts available then.
2. **Overlapping records:** random splitting can place related time windows or readings from the same machine lifecycle on both sides. Split by deployment time and, where appropriate, asset/lifecycle.
3. **Sales are censored demand:** observed sales are $\min(D, q)$. Low sales after running out do not prove low demand.

Concrete correction

A moving demand average for day t must use days before t. Record stockout flags and lost orders; our synthetic notebook observes full demand by design.

Backtest the entire decision policy

For each held-out day, predict using available information, choose feasible q_t, then observe D_t.

- Forecast: MAE for a point forecast; quantile loss/coverage for a quantile.
- Operation: mean cost, unserved units, stockout days, excess kits, capacity violations.
- Benchmark: constant inventory or an existing business rule; report both policies on the same days.
- Stress tests: higher shortage cost, less capacity, demand shift, and wider forecast errors.

A backtest estimates performance under its data and assumptions. It does not establish a causal field improvement or guarantee future cost savings.

Routing: what does a travel-time forecast change?

A vehicle chooses one of two routes before departure. These are toy scenarios:

Route	Time scenarios (min)	Mean	$\Pr(T > 30)$
A	20 (0.9), 60 (0.1)	24	0.1
B	26 (1.0)	26	0

For loss $T + \lambda \mathbf{1}\{T > 30\}$ (minute-equivalent units), expected losses are $24 + 0.1\lambda$ and 26.

- With $\lambda = 0$, choose A; with $\lambda > 20$, choose B.
- Mean time alone is insufficient for a lateness penalty.
- A fleet model also needs capacities, visit coverage, time windows, and feasible routes. ML predictions do not enforce those constraints.

Maintenance: a probability becomes an action

Let p be failure probability *during the next shift*. A toy model assumes service costs C_m and prevents that failure completely.

$$\text{No service: } pC_f, \qquad \text{Service: } C_m.$$

- Service if $p > C_m/C_f$. With $C_m = 100$, $C_f = 1000$, the threshold is 0.1, not automatically 0.5.
- If service reduces risk from p to rp, $0 \leq r < 1$, service only when $(1-r)pC_f > C_m$.
- Limited technicians turn independent choices into a constrained allocation problem.
- A classifier's near-failure score needs a calibrated probability for the correct horizon. Prediction alone does not prove a maintenance intervention works.

Lab 6 develops failure-risk classification and constrained maintenance choices.

How the three applications differ

	Inventory	Routing	Maintenance
Uncertain input	Demand distribution	Travel-time distribution	Shift failure probability
Decision	Prepared quantity	Route/vehicle assignment	Service selection and time
Cost tradeoff	Excess vs shortage	Travel vs late arrival	Service vs failure loss
Missing assumption to check	Lost sales observable?	Time depends on departure?	Service reduces risk?

The shared structure is data $\rightarrow$ uncertain input $\rightarrow$ feasible decision $\rightarrow$ measured outcome. The appropriate model follows the operational question.

Where decision-aware learning fits

- Standard pipeline: train a predictor for an error metric, then optimize with its output.
- Decision-aware learning: use the downstream objective to help choose or train the predictor.
- Our quantile loss already matches a particular inventory decision's asymmetric costs.
- Elmachtoub and Grigas develop an optimization-based loss for predicted objective coefficients; their numerical examples include shortest paths and portfolios.

Scope

This is a motivation for further study, not a claim that every optimizer can be differentiated through or that decision-aware training always wins.

Lecture companion: a 15–20 minute experiment

Open `notebooks/lecture_3_predict_then_optimize.ipynb`.

1. **Predict:** Will quantity 100 or 120 win on MAE and cost?
2. **Run:** Verify the exact three-scenario example and cost curve.
3. **Change:** Move shortage cost, demand noise, and capacity controls. Record the selected quantity and cost.
4. **Explain:** Fit a simple calendar predictor; calibrate residuals, select on validation, evaluate on untouched test days.

Lab 6 remains the longer forecasting/classification/blending/maintenance capstone. This companion isolates why a good prediction is useful only through the decision it supports.

Check your understanding

1. If $c_u = c_o$, which quantile minimizes expected mismatch loss?
2. Why can a capacity limit make the desired service target unattainable?
3. Why is tomorrow's confirmed promotion a valid input, but tomorrow's realized sales are not?
4. Why can a low-MAE forecast produce an expensive inventory policy?

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Answers

The median; feasible quantities may all be too small; only the promotion is known at commitment time; MAE weights shortage and excess equally, while operational costs may not.

Sources and further reading

- MIT OpenCourseWare, *Newsvendor Inventory Problem*, 15.772J, Fall 2014. Single-period model and critical-ratio background.
  MIT teaching notes
- Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed., Chapter 5: forecast evaluation and rolling-origin validation.
  Open textbook: The forecaster's toolbox
- Elmachtoub and Grigas, *Smart "Predict, then Optimize"*, Management Science 68(1), 2022, pp. 9–26.
  Authors' manuscript, arXiv:1710.08005
- Course Lab 6 reference snapshot: production-cell capstone.

The calculations and toy datasets here are reproducible in the companion notebook; they are not reported industrial outcomes.